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$$\vec{B}(s) = \vec{T}(s) \wedge \vec{N}(s) \quad \cdots\cdots\cdots (13)$$

So, the binormal of the curve $\alpha(s)$ is the constructed vector $B(s)$.

Finally, by (4), we have $B'(s) = \tau(s) N(s)$,

So the torsion of $\alpha(s)$ is precisely the prescribed function $\tau(s)$.

This proves existence and completes the fundamental theorem.

QED

Curve	Base point at $s=0$	Frenet tripod
original curve α	$p_0 = \alpha(0)$	$\{T_0, N_0, B_0\}$
Normalized curve $\beta = M\alpha$	$O = \beta(0)$	$\{e_1, e_2, e_3\}$

"End of lecture 3"

Lecture 4

The local canonical form

The key idea is to fix the Frenet tripod at one point and use it as a coordinate system for the nearby curve.

Let $\alpha: I \to \mathbb{R}^3$ be a curve parametrized by arc-length without singular points of order 1. k is a differentiable function with $k(s_0) > 0$. By continuity of k, we have $k > 0$ in a nbhd of s_0, so that the Frenet frame is defined there. Put $h = s - s_0$ and write $\tilde{\alpha}(h) = \alpha(s_0 + h) \quad \cdots\cdots (1)$

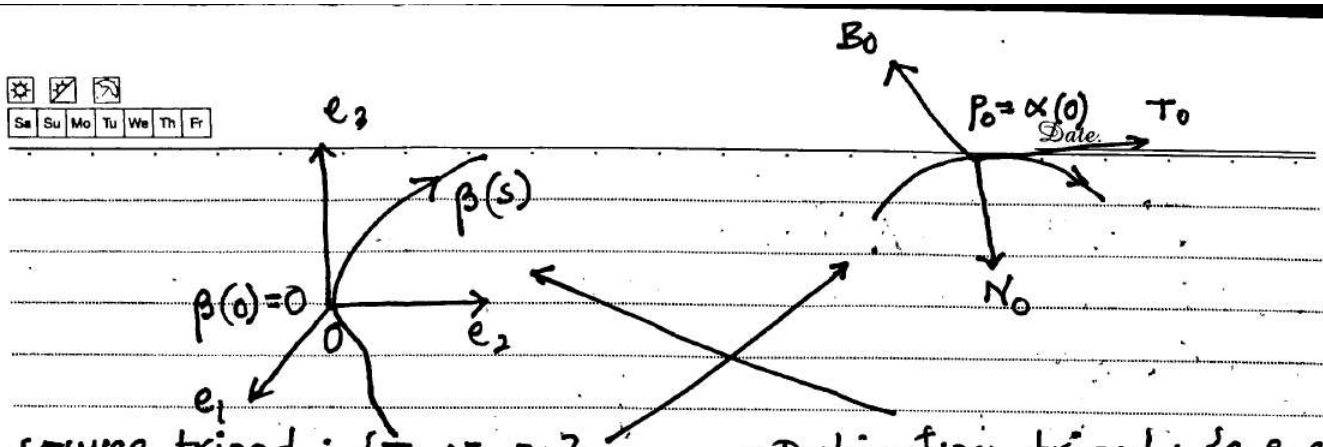
Then $\tilde{\alpha}'(h) = \alpha'(s_0+h) \frac{d}{dh}(s_0+h) = \alpha'(s_0+h) \quad \cdots\cdots (2)$

Thus $|\tilde{\alpha}'(h)| = 1$; [$\because |\alpha'(s)| = 1$ as α is arc-parametrized]

Write $p_0 = \alpha(0)$ (we dropped the tilde and we'll be denoting s by h from now on)

$T_0 = T(0)$, $N_0 = N(0)$, $B_0 = B(0) \quad \cdots\cdots\cdots (3)$

We place the spatial origin at p_0 and use T_0, N_0, B_0 as the positive x, y, z directions, resp.



- source tripod : $\{T_0, N_0, B_0\}$
  - located at $p_0 = \alpha(0)$
  - $\beta(s) = M(\alpha(s))$ passes through the origin
- Destination tripod : $\{e_1, e_2, e_3\}$
  - located at the origin O
  - $\alpha(s)$ passes through $\alpha(0) = p_0$

In the uniqueness proof: the source tripod $\{\bar{T}_0, \bar{N}_0, \bar{B}_0\}$ located at $\bar{p}_0 = \bar{\alpha}(s_0)$ was transported to the destination tripod $\{T_0, N_0, B_0\}$ located at $p = \alpha(s)$ with the rotation ρ represented by the matrix R with $R = F_0 \bar{F}_0^T$, where $F_0 = [T_0, N_0, B_0]$ and $\bar{F}_0 = [\bar{T}_0, \bar{N}_0, \bar{B}_0]$ being 3×3 matrices that are both orthogonal with unit determinant. Here, $\rho(\bar{T}_0) = T_0$, $\rho(\bar{N}_0) = N_0$, $\rho(\bar{B}_0) = B_0$. with $M(\bar{\alpha}(s_0)) = \alpha(s)$

In the present case, $M(p_0) = 0$, $\rho(T_0) = e_1$, $\rho(N_0) = e_2$, and $\rho(B_0) = e_3$ ..... (4)

And the rotation matrix $R = F_0^T$ simply with

$F_0 = [T_0, N_0, B_0]$ ..... (5)

For our present purpose, we denote the matrix F_0 by S so that $R = S^T$ with $S = [T_0, N_0, B_0]$ ..... (5)

We'll call $\beta(s)$ with $\beta(0) = 0$, the normalized curve defined by $\beta(s) = M(\alpha(s)) = \begin{bmatrix} x(s) \\ y(s) \\ z(s) \end{bmatrix}$ ..... (6)

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We first rotate about $P_0 = \alpha(0)$ and then translate it to the origin.

$R_{P_0}(x) = P_0 + \rho(x - P_0)$ ..... (7)

$\therefore \xi_1(s) = P_0 + \rho(\alpha(s) - P_0)$ (rotate about P_0)

$\beta(s) = \xi_1(s) - P_0$ (translate by $-P_0$)

$= P_0 + \rho(\alpha(s) - P_0) - P_0 = \rho(\alpha(s) - P_0)$ ..... (8)

Now, $x(s) = \langle \beta(s), e_1 \rangle$

$= \langle \rho(\alpha(s) - P_0), \rho(T_0) \rangle$: [using (4)]

$= \langle \alpha(s) - P_0, T_0 \rangle$: [Because ρ is orthogonal]

Similarly, $y(s) = \langle \alpha(s) - P_0, N_0 \rangle$, $z(s) = \langle \alpha(s) - P_0, B_0 \rangle$

Together $\begin{bmatrix} x(s) \\ y(s) \\ z(s) \end{bmatrix} = \begin{bmatrix} \langle \alpha(s) - P_0, T_0 \rangle \\ \langle \alpha(s) - P_0, N_0 \rangle \\ \langle \alpha(s) - P_0, B_0 \rangle \end{bmatrix}$ ..... (9)

We have, $\alpha'(s) = T(s)$, $\alpha''(s) = T'(s) = k(s)N(s)$ ..... (10)

$N'(s) = -\tau(s)B(s) - k(s)T(s)$ ..... (11)

Now, differentiate $\alpha''(s) = k(s)N(s)$ given by (10) to obtain

$\alpha'''(s) = \frac{d}{ds}(k(s)N(s)) = k'(s)N(s) + k(s)N'(s)$

$= k'(s)N(s) + k(s)[-\tau(s)B(s) - k(s)T(s)]$

$= -[k(s)]^2 T(s) + k'(s)N(s) - k(s)\tau(s)B(s)$

..... (11)

Set $k_0 = k(0)$, $k_0' = k'(0)$, $\tau_0 = \tau(0)$ ..... (12)

Evaluating after differentiating gives,

$$\boxed{\begin{aligned} \alpha'(0) &= T_0, \\ \alpha''(0) &= k_0 N_0, \\ \alpha'''(0) &= -k_0^2 T_0 + k_0' N_0 - k_0 \tau_0 B_0 \end{aligned}}$$ ..... (13)

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- **Taylor expansion**

$$\alpha(s) = \alpha(0) + s\alpha'(0) + \frac{s^2}{2}\alpha''(0) + \frac{s^3}{6}\alpha'''(0) + r_3(s)$$

Taylor remainder

with $\dfrac{|r_3(s)|}{|s|^3} \to 0$

Remainder calculation

$$\frac{1}{2}\int_0^s (s-u)^2 \alpha'''(u)\,du = \left[\frac{(s-u)^2}{2}\alpha''(u)\right]\Big|_0^s + \int_0^s (s-u)\alpha''(u)\,du$$

; [Integration by parts]

$$= -\frac{s^2}{2}\alpha''(0) + \left[(s-u)\alpha'(u)\right]\Big|_0^s + \int_0^s \alpha'(u)\,du$$

$$= -\frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - \alpha(0) = -\frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - P_0$$

..... (14)

[illegible]

Now, $r_3(s) = \dfrac{1}{6}\int_0^s (s-u)^3 \alpha^{(4)}(u)\,du$ ..... (15)

Now, $\dfrac{d}{du}\left[\dfrac{(s-u)^3}{6}\right] = -\dfrac{(s-u)^2}{2}$, ... (16)

i.e., $\dfrac{1}{6}d(s-u)^3 = \dfrac{(s-u)^2}{2}du$

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$$\therefore r_3(s) = \left[\frac{(s-u)^3}{6}\alpha'''(u)\right]\Big|_0^s + \frac{1}{2}\int_0^s (s-u)^2\alpha'''(u)\,du$$

$$= -\frac{s^3}{6}\alpha'''(0) - \frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - p_0$$

; [By using (4)]

Since $\alpha(0) = p_0$ ..... (17)

$$\therefore \alpha(s) = p_0 + s\alpha'(0) + \frac{s^2}{2}\alpha''(0) + \frac{s^3}{6}\alpha'''(0) + r_3(s) \quad \text{..... (18)}$$

with $r_3(s) = -\frac{s^3}{6}\alpha'''(0) - \frac{s^2}{2}\alpha''(0) - s\alpha'(0) + \alpha(s) - p_0$

Remainder estimate:

$$r_3(s) = \frac{1}{6}\int_0^s (s-u)^3\alpha^{(4)}(u)\,du$$

Substitute $u = sv$

u	v
0	0
s	1

Then $r_3(s) = \frac{s^4}{6}\int_0^1 (1-v)^3\alpha^{(4)}(sv)\,dv$

Since α is smooth, take C_4 to be a bound for $|\alpha^{(4)}|$ on a small closed interval about zero.

Now, $\|r_3(s)\| = \frac{|s|^4}{6}\left\|\int_0^1 (1-v)^3\alpha^{(4)}(sv)\,dv\right\|$

$$\leq \frac{|s|^4}{6}\int_0^1 \|(1-v)^3\alpha^{(4)}(sv)\|\,dv$$

$$= \frac{|s|^4}{6}\int_0^1 (1-v)^3\|\alpha^{(4)}(sv)\|\,dv$$

$$\leq \frac{C_4|s|^4}{6}\int_0^1 (1-v)^3\,dv = \frac{C_4|s|^4}{24} \quad \text{..... (19)}$$

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In particular, $\frac{\|r_3(s)\|}{|s|^3} \leq \frac{C_4 |s|}{24} \to 0$ ..... (20)

• Now (18) reads off as

$$\alpha(s) - p_0 = s\alpha'(0) + \frac{s^2}{2}\alpha''(0) + \frac{s^3}{6}\alpha'''(0) + r_3(s)$$

$$= sT_0 + \frac{s^2}{2}k_0 N_0 + \frac{s^3}{6}\left(-k_0^2 T_0 + k_0' N_0 - k_0 \tau_0 B_0\right) + r_3(s)$$

; [By applying (13)]

$$= \left(s - \frac{k_0^2 s^3}{6}\right) T_0 + \left(\frac{k_0}{2} s^2 + \frac{k_0'}{6} s^3\right) N_0 - \frac{k_0 \tau_0}{6} s^3 B_0 + r_3(s)$$

..... (21)

Apply ρ on (21) and then applying (8)

$$\beta(s) = \rho(\alpha(s) - p_0) = \left(s - \frac{k_0^2 s^3}{6}\right)\rho(T_0) + \left(\frac{k_0}{2} s^2 + \frac{k_0' s^3}{6}\right)\rho(N_0)$$

$$- \frac{k_0 \tau_0}{6} s^3 \rho(B_0) + \rho(r_3(s))$$

↕ Normalized curve

$$= \left(s - \frac{k_0^2 s^3}{6}\right) e_1 + \left(\frac{k_0}{2} s^2 + \frac{k_0' s^3}{6}\right) e_2$$

$$- \frac{k_0 \tau_0}{6} s^3 e_3 + \rho(r_3(s))$$

..... (22)

Define r_x, r_y, r_z to be the components of the rotated remainder

$$\rho\, r_3(s) = \underset{\text{numbers}}{r_x(s)}\, e_1 + r_y(s)\, e_2 + r_z(s)\, e_3 \quad \cdots (23)$$

$$r_x(s) = \langle \rho\, r_3(s), e_1 \rangle = \langle \rho(r_3(s)), \rho T_0 \rangle$$

$$= \langle r_3(s), T_0 \rangle$$

But $|r_x(s)| = |\langle r_3(s), T_0 \rangle| \leq \|\rho(r_3(s))\| = \|r_3(s)\|$

Similarly $|r_y(s)| \leq \|r_3(s)\|$

$|r_z(s)| \leq \|r_3(s)\|$

} – (24)

Hence, we can rewrite (22) as

$$\beta(s) = \left(s - \frac{k_0^2 s^3}{6} + r_x(s)\right) e_1 + \left(\frac{k_0}{2} s^2 + \frac{k_0'}{6} s^3 + r_y(s)\right) e_2 + \left(-\frac{k_0 \tau_0}{6} s^3 + r_z(s)\right) e_3$$

"canonical form"

From (6), one obtains

$$\begin{aligned} x(s) &= s - \frac{k_0^2}{6} s^3 + r_x(s) \\ y(s) &= \frac{k_0}{2} s^2 + \frac{k_0'}{6} s^3 + r_y(s) \quad , \quad |r_i(s)| \le C_4 \frac{|s|^4}{24} \\ z(s) &= -\frac{k_0 \tau_0}{6} s^3 + r_z(s) \qquad ; [\text{By } (24) \& (20)] \end{aligned} \qquad \cdots (25)$$

- Using the same remainders in the plane calculation

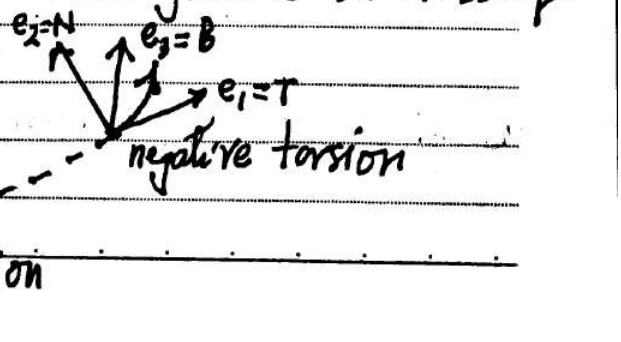
Recall that t-b plane is called the rectifying plane and the t-n plane is called the osculating plane.

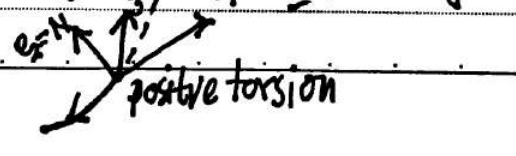
In the normalized coordinates (the frame being $\{e_1, e_2, e_3\}$) the rectifying plane is the $e_1 e_3$ plane, i.e., $y=0$ plane and the oscutating plane is the $e_1 e_2$ plane, i.e., $z=0$ plane. From (25), one then obtains,

$$\left.\begin{aligned} \frac{y(s)}{s^2} &= \frac{k_0}{2} + \frac{k_0'}{6} s + \frac{r_y(s)}{s^2} \to \frac{k_0}{2} > 0 \\ \frac{z(s)}{s^3} &= -\frac{k_0 \tau_0}{6} + \frac{r_z(s)}{s^3} \to -\frac{k_0 \tau_0}{6} \end{aligned}\right\} \cdots (26)$$

Thus $y(s) > 0$ for all sufficiently small $s \neq 0$, on both sides of zero. If $\tau_0 \neq 0$, $z(s)$ changes sign at $s=0$ because s^3 changes sign. In do Carmo's convention, negative torsion gives a crossing toward the positive binormal side

And positive torsion gives a crossing toward the negative binormal side.

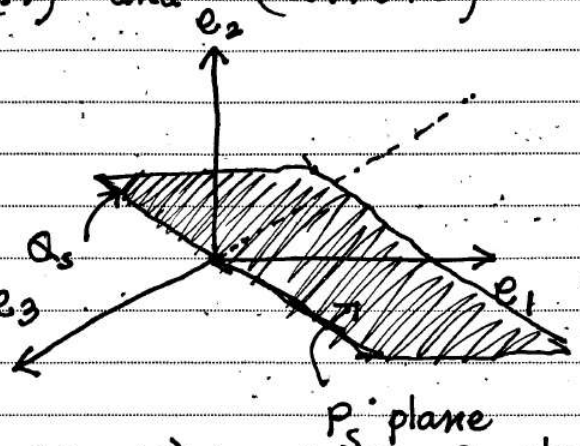




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• In our normalized coordinates, the tangent line is the X-axis; ~~the~~, i.e., e_1. The plane P_s contains $\beta(s) = (x(s), y(s), z(s))$ and $(x(s), 0, 0)$



The line $Q_s = (0, y(s), z(s))$ lies in the P_s plane. But since its X-coordinate is 0, the line lies in the YZ plane, i.e., The plane P_s intersects the YZ plane at the line Q_s. In fact, the P_s plane is the one that contains the tangent vector e_1 at the origin and a nearby point $\beta(s)$ of the normalized curve.

Now, $y(s) > 0$ for small $s \neq 0$. Dividing Q_s by $y(s)$ changes its length, not its direction.

Thus $$P_s = \text{Span}\left\{ e_1, \left(0, 1, \frac{z(s)}{y(s)}\right) \right\} \quad \cdots\cdots (27)$$

So, looking along the x-axis, the cross-section of P_s is the line Q_s with slope $$c(s) = \frac{z(s)}{y(s)} \quad \cdots\cdots (28)$$

The Canonical form (28) gives

$$\frac{y(s)}{s^2} \to \frac{k_0}{2} > 0, \quad \frac{z(s)}{s^3} \to -\frac{k_0 \tau_0}{6}$$

as given by (26). Therefore, $$c(s) = \frac{z(s)}{y(s)} = s \frac{z(s)/s^3}{y(s)/s^2} \quad \cdots\cdots (29)$$

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From (26), one obtains $\frac{z(s)/s^3}{y(s)/s^2} \to -\frac{T_0}{3}$ as $s \to 0$

so that $c(s) \to 0$ as $s \to 0$; [From (29)]

Hence, the second direction $(0, 1, c(s))$ in (27) tends to e_2, while the first direction is always e_1. The limiting plane is therefore the XY plane, $z = 0$, or the osculating plane.

"End of lecture 4"

- Next lecture: global properties of plane curves.